



ADVANCED GEOPHYSICS INSTRUMENTS  
WWW.MAE-SRL.IT

USER HANDBOOK

# DataLogger Live Dashboard

Complete User Guide — Monitor v2

VMR624

DL8

DOCUMENT TYPE

**User  
Handbook**

VERSION

**Monitor v2**

LANGUAGE

**English**

PLATFORM

**Web — All  
Browsers**

DASHBOARD URL

[maeservice.it/beta](https://www.maeservice.it/beta)

YEAR

**2026**

# TABLE OF CONTENTS

|           |                                                 |           |
|-----------|-------------------------------------------------|-----------|
| <b>1</b>  | <b>General Information</b>                      | <b>3</b>  |
|           | 1.1 Introduction                                | 3         |
|           | 1.2 Intended Use                                | 3         |
|           | 1.3 Safety & Icons                              | 3         |
| <b>2</b>  | <b>Getting Started — Access &amp; Login</b>     | <b>4</b>  |
|           | 2.1 System Requirements                         | 4         |
|           | 2.2 Login Procedure                             | 4         |
|           | 2.3 Logout                                      | 4         |
| <b>3</b>  | <b>Works Page</b>                               | <b>5</b>  |
|           | 3.1 Selecting a Work / Project                  | 5         |
|           | 3.2 Work Status & Summary Panel                 | 5         |
| <b>4</b>  | <b>Dashboard Layout Overview</b>                | <b>6</b>  |
|           | 4.1 Top Navigation Bar                          | 6         |
|           | 4.2 Left Sidebar                                | 6         |
|           | 4.3 Main Content Area                           | 7         |
| <b>5</b>  | <b>Period, Channel &amp; Interval Filters</b>   | <b>8</b>  |
| <b>6</b>  | <b>Main Chart</b>                               | <b>9</b>  |
|           | 6.1 Single Channel View                         | 9         |
|           | 6.2 All Channels Overlay View                   | 9         |
|           | 6.3 Interactive Features                        | 9         |
| <b>7</b>  | <b>Live Mode &amp; Auto-Refresh</b>             | <b>10</b> |
| <b>8</b>  | <b>Device Management</b>                        | <b>11</b> |
|           | 8.1 Device List & Status LED                    | 11        |
|           | 8.2 Device Information Card                     | 11        |
| <b>9</b>  | <b>KPI Cards &amp; Statistics</b>               | <b>12</b> |
| <b>10</b> | <b>Secondary Charts &amp; Measurement Table</b> | <b>13</b> |

|           |                                  |    |
|-----------|----------------------------------|----|
| <b>11</b> | Alerts & Threshold Settings      | 14 |
| <b>12</b> | Export & File Download           | 15 |
| <b>13</b> | Map & Device Location            | 17 |
| <b>14</b> | Power Supply Monitor             | 18 |
| <b>15</b> | Device Types & Channel Reference | 19 |
| <b>16</b> | Language & Theme Settings        | 20 |
| <b>17</b> | Troubleshooting                  | 21 |

## 1.1 Introduction

The **MAE DataLogger Live Dashboard** is a web-based monitoring platform for geophysical instruments manufactured by MAE s.r.l. It provides live access to measurement data transmitted by field devices, including vibration monitors (VMR624) and data loggers (DL8).

Operators and technical staff can view, analyse and export data directly in any modern web browser — no software installation required. The interface is responsive and works on desktop computers, tablets and mobile devices.

### LIVE DASHBOARD URL

The dashboard is accessible at: **<https://www.maeservice.it/beta/>**  
An active account with valid credentials provided by MAE is required to access the system.

## 1.2 Intended Use

This platform is designed for qualified technical operators trained by MAE or an authorised representative. It is intended for:

- Real-time monitoring of field measurement devices
- Visualisation and analysis of recorded geophysical data
- Reviewing events and threshold violations recorded by the devices
- Exporting and archiving measurement data in standard formats
- Remote supervision of monitoring sites with multiple devices

### ATTENTION

Any use of the dashboard for purposes other than those described above is considered improper use. MAE s.r.l. assumes no liability for damages resulting from unauthorised or improper use of this software.

The technical operator is solely responsible for the correctness and interpretation of the data produced.

## 1.3 Icons Used in This Handbook

| Icon / Label                                                                                       | Meaning                                                                           |
|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|  <b>NOTE</b>      | Informational note — read carefully to understand correct operation.              |
|  <b>ATTENTION</b> | Warning message regarding safety or correct configuration of the system.          |
|  <b>IMPORTANT</b> | Critical instruction — failure to follow may result in data loss or system error. |
|  <b>TIP</b>       | Useful hint to improve efficiency or user experience.                             |

# 02 Getting Started — Access & Login

## 2.1 System Requirements

| Requirement         | Specification                                                          |
|---------------------|------------------------------------------------------------------------|
| Browser             | Google Chrome 90+, Mozilla Firefox 88+, Microsoft Edge 90+, Safari 14+ |
| Internet Connection | Required — the dashboard communicates with the MAE cloud service       |
| Screen Resolution   | Minimum 1024 × 768 px; recommended 1440 × 900 px or higher             |
| Mobile / Tablet     | Fully supported — responsive layout adapts to all screen sizes         |
| Local Software      | None required — fully browser-based application                        |
| Credentials         | Username and password provided by MAE s.r.l.                           |

## 2.2 Login Procedure

### 1 Open the Dashboard

Navigate to

**<https://www.maeservice.it/beta/>**

in your browser. You will be directed to the Works page with the login form visible.

### 2 Enter Credentials

Enter your

**Username**

in the first field. Press **Tab** or **Enter** to move to the

## Password

field and enter your password.

3

### Sign In

Click the

### Sign In

button or press . Your credentials are verified securely against the MAE authentication server.

4

### Session Saved

On successful login, your session is stored locally in the browser. You will remain logged in across page refreshes until you explicitly log out.

## 2.3 Logout

To log out, click the **Logout** button in the top-right corner of the navigation bar (shown as a door/exit icon). This clears your session and returns you to the login page.

### ATTENTION — SHARED COMPUTERS

Always log out when using the dashboard on a shared or public computer. Simply closing the browser tab does **not** end your session — always use the Logout button.

The **Works Page** is the first screen you see after logging in. It lists all monitoring sites associated with your account. Each site — referred to as a *work* — represents a specific deployment, such as a construction project, a geological survey or a monitoring campaign.

## 3.1 Selecting a Work / Project

### 1 Browse Available Works

Your works are displayed as cards in a grid. Each card shows the work name, location, number of devices and whether the work is active or inactive.

### 2 Search or Filter

Use the

**search bar**

at the top to find a work by name or location. Use the filter chips (

**All**

,

**Active**

,

**Inactive**

) to narrow the list by status.

### 3 Open the Dashboard

Click

**Open Dashboard**

on the desired work card. The dashboard will load showing all devices linked to that work.

## 3.2 Work Status & Summary Panel



A **Summary Panel** at the top of the Works page shows aggregated statistics across all your works:

| Indicator | Description                                           |
|-----------|-------------------------------------------------------|
| Active    | Number of works currently in active monitoring status |
| Inactive  | Number of works that are inactive or suspended        |
| Total     | Total number of works in your account                 |
| Devices   | Total number of devices across all works              |

Each work card includes an LED indicator: **green** = at least one device connected recently, **orange** = devices connected but not recently, **grey** = no devices or no recent connection.

# 04 Dashboard Layout Overview

The dashboard is organised into three areas: the **top navigation bar**, the **left sidebar** and the **main content area**. Familiarising yourself with this layout will help you navigate the platform efficiently.

## 4.1 Top Navigation Bar

The top bar spans the full width of the screen and contains controls that apply globally to the current session:

| Control                   | Function                                                                                                                           |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>MAE Logo / Brand</b>   | Displays the application name "DataLogger — MONITOR v2" and the active work subtitle (work name, location, device count).          |
| <b>Period From / To</b>   | Date range selector. Defines the start and end dates for data retrieval. Changing these dates pauses Live Mode.                    |
| <b>Apply Button</b>       | Applies the selected date range and reloads data from the server for the chosen period.                                            |
| <b>Channel Selector</b>   | Drop-down to select which measurement channel to display in the main chart. Available channels depend on the device configuration. |
| <b>Interval Selector</b>  | Quick filter: All, Last 1h, Last 3h, Last 6h, Last 12h, Last 24h. Filters the loaded dataset without re-executing the search.      |
| <b>Live / Paused Pill</b> | Status indicator. LIVE (green) = auto-refreshing every 120 s. PAUSED (orange) = manual mode, click to resume live.                 |
| <b>Refresh Countdown</b>  | Shows the seconds remaining until the next automatic data refresh. Click to force an immediate refresh.                            |

| Control         | Function                                                                              |
|-----------------|---------------------------------------------------------------------------------------|
| Language Toggle | Switch the interface language between EN (English) and IT (Italian).                  |
| Theme Toggle    | Switch between Dark and Light visual themes.                                          |
| Alerts Button   | Opens the active alerts panel. A red badge shows the number of unacknowledged alerts. |
| Export Button   | Opens the export modal with options for CSV, JSON, Print/PDF, and raw file download.  |
| Logout Button   | Logs out the current user and returns to the login page.                              |

## 4.2 Left Sidebar

### NAVIGATION

- Works — Return to the Works selection page
- Alerts — Open the alerts panel (shows badge count when active)
- Export — Open the export modal

### DEVICE LIST

- Lists all devices in the active work
- Shows connection status LED for each device
- Shows device type, position name, and serial number
- Click any device to switch to it and reload its data

## 4.3 Main Content Area

| Zone         | Content                                                                                   |
|--------------|-------------------------------------------------------------------------------------------|
| Alert Banner | Appears automatically when threshold violations are detected. Dismissible per alert.      |
| KPI Cards    | One card per measurement channel showing the latest value, unit, and status colour.       |
| Main Chart   | Interactive time-series chart for the selected channel (or all channels in overlay mode). |

| Zone                  | Content                                                                                    |
|-----------------------|--------------------------------------------------------------------------------------------|
| Secondary Mini-Charts | Compact charts for all non-primary channels, shown simultaneously for comparison.          |
| Measurement Log Table | Tabular view of all records in the current filtered dataset.                               |
| Device Info Card      | Technical details of the active device: battery, signal, GPS, last connection, IP, memory. |

# 05 Period, Channel & Interval Filters

## Date Range (Period)

Use the **From** and **To** date fields in the top bar to set the time period for which data is displayed. The system loads up to **150 records** for the selected period.

- By default, both dates are set to **today** (or the date of the device's most recent transmission).
- Changing either date field automatically **pauses Live Mode**.
- After setting the dates, click **Apply** to load data for the new range.
- To return to live monitoring, click the **PAUSED** pill — this resets the dates to today and resumes automatic updates.

### NOTE — DATA LIMIT

Up to **150 records** are shown per request. If a device transmits data very frequently, only the most recent 150 samples within the selected period will be visible. Use the Export function to download the full dataset.

## Channel Selector

The **Channel** drop-down controls which measurement channel is shown in the main chart. The available channels depend on the connected device and the site configuration. Selecting **"All Channels"** switches to overlay mode, displaying all channels on the same chart at once.

## Interval Quick-Filter

The **Interval** drop-down lets you focus on a recent time window within the data already loaded, without fetching new data from the server:

| Option        | Effect                                                 |
|---------------|--------------------------------------------------------|
| All           | Display all records loaded for the selected date range |
| Last 1 hour   | Show only records from the last 60 minutes             |
| Last 3 hours  | Show only records from the last 3 hours                |
| Last 6 hours  | Show only records from the last 6 hours                |
| Last 12 hours | Show only records from the last 12 hours               |
| Last 24 hours | Show only records from the last 24 hours               |

Changing the interval updates the chart and table instantly. No new data is fetched from the server.

The main chart is the central element of the dashboard. It displays a smooth time-series plot of the selected measurement channel, rendered at high quality on any screen size.

## 6.1 Single Channel View

When a single channel is selected, the chart shows:

- A smooth curve connecting all data points, with a gradient fill below the line
- Value labels on the left axis (minimum, midpoint and maximum)
- Time labels along the bottom axis
- Light horizontal grid lines to aid value reading
- A statistics bar below the chart showing: MIN, MAX, AVG, RANGE and total SAMPLES
- The channel name and unit of measurement in the chart header

## 6.2 All Channels Overlay View

Select **"All Channels"** to overlay all measurement channels on the same chart. Each channel is normalised to a 0–100% scale so they can be compared visually regardless of their units:

- Each channel is shown in a distinct colour
- A colour-coded legend below the chart lists each channel's name and min–max range
- The vertical axis shows percentages (0%, 50%, 100%)
- Hovering over the chart shows the values of all channels at that point in time

### NOTE

Y-axis zoom is not available in All Channels mode. Switch to a single channel to use the zoom feature.

## 6.3 Interactive Features

### HOVER TOOLTIP

Move the mouse cursor over the chart area to see a tooltip showing the **date**, **time**, and **exact value** at the nearest data point. A vertical indicator line follows the cursor.

### Y-AXIS ZOOM

**Scroll up** with the mouse wheel to zoom in on the Y axis (magnify the value range). **Scroll down** to zoom out. Zoom range: 0.05× to 10×. The hint text "scroll to zoom" is shown in the chart corner.

### RESET ZOOM

**Double-click** anywhere on the chart to reset the Y-axis zoom to the default (1×). The hint changes to "🔄 dbclick to reset" when zoom is active.

### STATS SUMMARY

Below the chart, five statistics are always visible for the current view: **MIN**, **MAX** (highlighted in channel colour), **AVG**, **RANGE**, and **SAMPLES** (total record count).



# 07 Live Mode & Auto-Refresh

Live Mode keeps the dashboard up to date by automatically refreshing measurement data at regular intervals. It is the recommended mode when monitoring active devices in the field.

## Behaviour Overview

| State         | Behaviour                                                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LIVE</b>   | Data is refreshed every <b>120 seconds</b> . The countdown timer counts down from 120 and resets after each refresh. The date range is locked to today. |
| <b>PAUSED</b> | No automatic refresh. The operator controls when data is reloaded using the Apply button or by resuming Live Mode.                                      |

## Automatic State Detection

When the dashboard opens or when you switch between devices, the system automatically decides whether Live Mode should be active:

- If the device transmitted data **today** → Live Mode starts automatically
- If the last transmission was on a **previous date** → Live Mode is paused and the date range is set to the most recent data available
- If **no data** is found for the current period → an error message is shown and Live Mode remains off

## Manual Control

1

### Pause Live Mode

Change the date range using the date pickers and click **Apply**

. Live Mode pauses automatically when a custom date range is applied.

2

## Resume Live Mode

Click the

**PAUSED**

status pill. The dates reset to today, data is immediately refreshed, and the 120-second countdown restarts.

3

## Force Immediate Refresh

Click the countdown timer button at any time to trigger an immediate data refresh without waiting for the countdown to reach zero.

### TIP

Data refreshes every 120 seconds automatically. To get the latest data immediately, click the countdown button at any time. For reviewing historical data, pause Live Mode and set the date range manually.

## 8.1 Device List & Status LED

All devices belonging to the active work are listed in the left sidebar. Each entry shows:

- A **coloured status LED** indicating connection freshness
- The **device type** (e.g., VMR624, DL8)
- The **position/location name** (e.g., "PULVINO C13/C14")
- The **serial number**

| LED Colour | Meaning                           | Time Since Last Connection |
|------------|-----------------------------------|----------------------------|
| ● Green    | Online — recently connected       | Within the last 15 minutes |
| ● Orange   | Warning — not connected recently  | 15 to 30 minutes ago       |
| ● Red      | Offline — no recent connection    | More than 30 minutes ago   |
| ● Grey     | Unknown — no connection timestamp | No data available          |

Click any device in the list to make it the **active device**. The dashboard will reload all data for that device using the current date range. The selected device is highlighted in blue.

### ATTENTION — DEVICE SWITCH

When switching devices, all data on screen is cleared immediately to avoid showing outdated information. Data for the newly selected device is loaded before the chart and table are updated.

## 8.2 Device Information Card

The Device Information Card appears in the main content area and shows the technical details of the active device. It is refreshed automatically each time data is updated:

| Field                   | Description                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------------|
| Serial Number           | Unique hardware identifier of the device (matricola)                                          |
| Type / Typology         | Device model (e.g., VMR624, DL8)                                                              |
| Last Connection         | Timestamp of the most recent server contact                                                   |
| Last Diagnostic         | Timestamp of the most recent data transmission                                                |
| Signal Strength         | GSM signal level (0–4 bars); colour-coded badge                                               |
| Memory Free             | Available SD card space in GB                                                                 |
| Battery Voltage         | Current battery voltage in Volts                                                              |
| USB Voltage             | Current USB supply voltage in Volts                                                           |
| AUX Voltage             | Current auxiliary supply voltage in Volts                                                     |
| Location / Work Place   | Name of the monitoring site or work location                                                  |
| Position / Device Place | Specific installation point of this device                                                    |
| IP Address              | Local network IP address and port                                                             |
| Public IP               | Public IP address and port (if available)                                                     |
| Coordinates             | GPS latitude and longitude (decimal degrees)                                                  |
| Mini Map                | Clickable thumbnail map showing device location (only if valid GPS coordinates are available) |

**DEVICE TIME WARNING**

If the device's internal clock is not synchronised with the server time, a warning banner is displayed: **"Warning! Device time is incorrect!"**

— This may indicate GPS/NTP sync issues on the device. Contact MAE technical support if this warning persists.

# 09 KPI Cards & Statistics

**KPI Cards** appear in a row above the main chart — one for each measurement channel of the active device. They give a quick summary of the latest reading and status for every channel.

## Card Contents

| Element                | Description                                                                                     |
|------------------------|-------------------------------------------------------------------------------------------------|
| <b>Channel Label</b>   | Human-readable name of the measurement channel                                                  |
| <b>Current Value</b>   | The most recent recorded value for this channel, displayed in large text with unit              |
| <b>Trend Indicator</b> | ▲ increasing, ▼ decreasing, or ● stable — based on comparison with the previous record          |
| <b>Status Colour</b>   | Green = within normal range, Amber = warning threshold exceeded, Red = alert threshold exceeded |
| <b>Alarm Badge</b>     | Shown if this channel has an active event/alarm from event file detection                       |

### NOTE — KPI VISIBILITY

The KPI cards are hidden when no data is available for the current date range. They reappear automatically once data is successfully loaded. When a threshold is exceeded, the corresponding KPI card border is highlighted in the alert colour.

## Colour Coding

**NORMAL****WARNING****ALERT / CRITICAL**

Value is within  
configured safe  
range. No action  
required.

Value has exceeded  
a warning threshold.  
Attention  
recommended.

Value has exceeded  
the critical  
threshold.  
Immediate review  
required.

# 10 Secondary Charts & Measurement Table

## Secondary Mini-Charts

A row of compact **mini-charts** appears below the main chart — one for each channel that has data. They let you compare all channels side by side without leaving the main view.

Each mini-chart shows:

- The channel name and an in-line **min** → **max range** value indicator
- A compact area chart with a coloured gradient fill
- The same time axis as the main chart

Mini-charts are for visual reference only — they do not support zoom or hover interaction. They refresh automatically alongside the main chart.

## Measurement Log Table

The measurement table lists all records in the current dataset in chronological order. Columns are adapted to match the active device type:

| Column            | Description                                                         |
|-------------------|---------------------------------------------------------------------|
| Date              | Recording date in DD/MM/YYYY format                                 |
| Time              | Recording time in HH:MM:SS format                                   |
| Channel 1...<br>N | One column per measurement channel with the recorded value and unit |
| Status            | Optional status column (device-specific)                            |

A **record count** is shown below the table header (e.g., "150 records"). Use the **search box** to filter rows by any value — useful for quickly locating a specific date or reading.



**TIP — QUICK EXPORT**

A quick **Export CSV** button is available in the table footer, providing one-click export of the currently displayed filtered dataset without opening the full Export modal.

## 11

# Alerts & Threshold Settings

The alert system checks measurement values against configured thresholds each time data is refreshed. When a threshold is exceeded, the operator is notified immediately on screen.

## Alert Types

### Warning Alert

Value has crossed a warning threshold. Attention is recommended but not necessarily urgent.

### Critical Alert

Value has crossed a critical threshold. Immediate inspection is required.

Active alerts are shown as coloured banners above the KPI cards. Each banner displays the channel name, the current value, the threshold that was exceeded and when the violation occurred. You can dismiss individual alerts by clicking the **×** button on the banner.

## Configuring Thresholds

Click the **Thresholds** icon in the sidebar to open the configuration panel. The following parameters can be set for each channel:

| Parameter           | Description                                                                       |
|---------------------|-----------------------------------------------------------------------------------|
| <b>Min Warning</b>  | Lower warning level — alert triggered when value falls below this point           |
| <b>Max Warning</b>  | Upper warning level — alert triggered when value rises above this point           |
| <b>Min Critical</b> | Lower critical level — critical alert triggered when value falls below this point |

| Parameter           | Description                                                                       |
|---------------------|-----------------------------------------------------------------------------------|
| Max Critical        | Upper critical level — critical alert triggered when value rises above this point |
| Enable Alerts       | Master toggle to enable or disable the alert evaluation for all channels          |
| Sound Notifications | Toggle to enable an audible beep when a new alert is triggered                    |

#### ATTENTION — THRESHOLD PERSISTENCE

Threshold values are stored in the browser's local storage. They are preserved across sessions on the same browser and device, but are **not synchronised** between different computers or browsers. If you use the dashboard from multiple devices, configure thresholds separately on each.

## Event-Based Alarms (VMR624)

The VMR624 generates event files when a vibration threshold is exceeded on the device itself. When such a file is detected for the selected period, the corresponding KPI card is marked with an alarm badge and the event appears in the Alarms Panel (accessible via the Alerts button in the sidebar). Clicking an alarm entry opens a detail view with the full event data.

# 12 Export & File Download

Click the **Export** button in the top bar to open the export panel. Four export formats are available. All exports are based on the **data currently displayed** (i.e. the active date range and interval filter), unless stated otherwise.

## 12.1 CSV Export

Exports all records in the current filtered view as a **Comma-Separated Values** file, compatible with Microsoft Excel, Google Sheets, and any standard data analysis tool.

- **Filename:** `datalogger_{deviceId}_export.csv`
- **Content:** Header row with column names and units, followed by one data row per record
- **Columns:** Date, Time, and one column per measurement channel (values to 3 decimal places)
- **Encoding:** UTF-8

## 12.2 JSON Export

Exports the full dataset as a structured **JSON** file, suitable for system integration, data import, or custom data processing scripts.

- **Filename:** `datalogger_{deviceId}_export.json`
- **Structure:**

```
{ "device": "...", "exported": "...", "records": [...] }
```
- **Records array:** Each element contains date, time, and all channel values

## 12.3 Print / PDF Report

Generates a comprehensive printable HTML report in a new browser tab. The report can be saved as a PDF using the browser's built-in print dialog (`Ctrl + P` → Save as PDF).

| Report Section     | Content                                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|
| Header             | MAE logo, device serial number and type, date range, export timestamp                                                 |
| Device Information | 9 technical fields in a 3×3 grid: serial, type, last connection, battery, USB, AUX voltages, signal, memory, location |
| Data Records Table | Full tabular listing of all records in the current filtered view, with date, time, and all channel values             |
| Data Visualisation | Mini charts for each channel, plus a combined overview chart                                                          |
| Footer             | MAE branding, generation timestamp, and website reference                                                             |

## 12.4 Device File Download

This option opens a file browser where you can view and download raw recordings stored on the MAE server. It is useful for retrieving the original files for detailed offline analysis.

How to use the file browser:

- Set a date range and choose the file type (Event, Circular or Daily)
- Click **Search Files** to retrieve the list of matching files
- Files are shown in a table with their name, timestamp and type — click **Download** to save any file locally
- Use the ← / → buttons to navigate through results when more than 50 files are available

| File Type | Description                                                                               |
|-----------|-------------------------------------------------------------------------------------------|
| Event     | Event files — triggered recordings when a measurement threshold is exceeded on the device |

| File Type | Description                                                      |
|-----------|------------------------------------------------------------------|
| Circular  | Circular buffer files — continuous recordings in rotating format |
| Daily     | Daily log files — full day recordings archived at midnight       |

# 13 Map & Device Location

The map shows the geographic location of the active device. Open it by clicking the **map thumbnail** in the Device Information card.

## Map Controls

| Control             | Function                                                                                |
|---------------------|-----------------------------------------------------------------------------------------|
| Zoom In / Out       | Buttons to increase or decrease the map zoom level                                      |
| Re-centre           | Returns the map view to the active device's location                                    |
| Map Style Selector  | Switch between three base map tiles: <b>Streets</b> , <b>Satellite</b> , <b>Terrain</b> |
| Quick Jump / Search | Type a city or address in the search box to navigate the map to that location           |

## Device Sidebar (in Map Modal)

The left panel lists all devices in the active work. Each entry shows the connection status, device name, serial number, type and GPS coordinates. Clicking a device centres the map on its location and updates the information panel.

### NOTE — GPS AVAILABILITY

The map is only displayed if the device has valid GPS coordinates. Coordinates of **0.000000°**, **0.000000°** are treated as invalid and no map is shown. If the device does not have a GPS module or has not yet acquired a fix, the map preview and modal will not be available.

# 14 Power Supply Monitor

The **Power Supply Monitor** shows the voltage history for the device's power inputs. Open it by clicking the battery/power icon in the Device Information card.

## Monitored Channels

### BATTERY

Internal battery voltage. Nominal range: **3.7 V – 4.2 V** (DL8) or **10.0 V – 12.5 V** (VMR624). Charge status indicator shows if the battery is currently charging.

### USB SUPPLY

USB input voltage. Nominal value: **5.0 V**. Range 4.8 V – 5.0 V is considered healthy.

### AUX SUPPLY

Auxiliary external power supply voltage. Nominal range: **12.0 V – 15.0 V**. Represents the external DC supply when connected.

## Power Monitor Features

- **KPI bars** — Shows the current voltage value for each channel with a colour-coded status bar (green = nominal, amber = low, red = critical)
- **Combined chart** — All three voltage channels overlaid on a single time-series chart for trend comparison
- **Individual sparklines** — Compact dedicated chart per channel with min/max range annotation
- **Channel visibility toggle** — Show or hide individual channels in the combined chart
- **Charge status** — Green indicator when the battery is actively charging, orange when discharging
- **Last update timestamp** — Shows when the power data was last refreshed from the server



#### NOTE — DATA HISTORY

The power monitor stores the last **10 diagnostic records** when the modal is closed and the last **100 records** when the modal is open. Updates are synchronised with the main dashboard auto-refresh interval (120 seconds).

# 15 Device Types & Channel Reference

The dashboard supports the device types listed below. Measurement channels are configured dynamically for each installation — the channel names and units shown in the dashboard reflect the specific setup agreed with the client.

## 15.1 VMR624 — Vibration Monitor

The VMR624 monitors vibration at a measurement point. Its channels typically cover displacement, frequency, temperature and supply voltage, with names and units defined per installation.

The VMR624 supports **event detection**: when a vibration threshold is exceeded, the device generates an event file and uploads it to the server. The dashboard detects these files automatically and displays an alarm on the relevant channel.

## 15.2 DL8 — Data Logger

The DL8 is a multi-channel data logger used primarily for settlement and tilt monitoring. It can be connected to displacement sensors, tilt sensors or other instruments, with channel names and units defined per installation.

The DL8 does **not** generate event files. Threshold violations are detected and displayed by the dashboard based on the thresholds configured in the alert settings.

## Dynamic Channel Configuration

Channel names, units and the number of channels are provided directly by the device and can vary between installations. This means the dashboard

adapts automatically to each site's setup without requiring any software changes.

# 16 Language & Theme Settings

## Language Selection

The dashboard supports two interface languages. Use the language toggle buttons in the top navigation bar to switch:

### EN — ENGLISH

Default language. All labels, buttons, error messages, and tooltips are displayed in English.

### IT — ITALIAN

Full Italian translation available. Click **IT** to switch. The language selection is remembered for the current session.

The selected language is applied instantly across the entire interface — including chart labels, KPI titles, status messages and error text.

## Visual Theme

Use the sun/moon icon in the top bar to toggle between visual themes:

### DARK MODE

Deep navy background with high-contrast coloured accents. Ideal for low-light environments and extended monitoring sessions.

**Default theme.**

### LIGHT MODE

Clean white background with dark text. Ideal for bright office environments, printing, and screen sharing.

The theme preference is stored in the browser and persists across sessions.

## 17

## CHAPTER SEVENTEEN

# Troubleshooting

| Problem                                   | Cause & Solution                                                                                                                                                                                                           |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Login fails — "Invalid credentials"       | Verify username and password. Ensure Caps Lock is off. Contact MAE support if credentials were recently changed.                                                                                                           |
| Dashboard redirects to Works page on load | Session has expired or no work is selected. Log in again and select a work from the Works page.                                                                                                                            |
| "No devices found" message                | No devices are assigned to the selected work, or the server connection failed. Verify internet connectivity and contact MAE if the issue persists.                                                                         |
| "No data for this period"                 | The device has no recorded data for the selected date range. Try expanding the date range or switch to a date when the device was active.                                                                                  |
| Error message shown in red banner         | The server returned an error. Possible causes: expired session (log out and back in), no internet connection, or temporary server maintenance. Note the error code shown and contact MAE support if the problem continues. |
| Chart is empty                            | The selected date range may not contain data for this device. Check that the Interval filter is not set to a window too narrow for the available records — try setting it to "All".                                        |
| Live Mode does not start automatically    | The device has not transmitted data today — it may be offline. Check the status LED in the sidebar. You can still click PAUSED to manually resume Live Mode when the device comes back online.                             |
| "Warning! Device time is incorrect!"      | The device's internal clock does not match the server time. GPS/NTP sync may have failed. Check the device in the field or contact MAE technical support.                                                                  |

| Problem                                       | Cause & Solution                                                                                                                                                 |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Map is not shown in device info card</b>   | The device does not have valid GPS coordinates. Coordinates 0.0000°, 0.0000° are treated as invalid. Ensure the device has GPS enabled and has acquired a fix.   |
| <b>Export file is empty</b>                   | There is no data in the current view. Make sure the chart is showing data before exporting — set the Interval filter to "All" and try again.                     |
| <b>Page is slow or unresponsive</b>           | Reduce the date range to a shorter period. Large datasets can slow down the browser. Closing other open browser tabs may also help.                              |
| <b>Thresholds reset after closing browser</b> | Threshold settings are stored in browser local storage. They may be cleared if browser history/cache is deleted. Re-enter thresholds after a browser data clear. |

#### TECHNICAL SUPPORT

For issues not resolved by the steps above, contact MAE s.r.l. technical support:

Website: [www.mae-srl.it](http://www.mae-srl.it) | Dashboard: <https://www.maeservice.it/beta/>

Please provide the device serial number, the error message displayed, and a description of the steps taken when the problem occurred.



## **MAE s.r.l. — Advanced Geophysics Instruments**

Zona Industriale Fresilia — 86095 Frosolone (IS) — Italy

[www.mae-srl.it](http://www.mae-srl.it) | <https://www.maeservice.it/beta/>

This document is intended for authorised users of the MAE DataLogger Live Dashboard system.

All content is subject to change without notice. © 2026 MAE s.r.l. All rights reserved.